

THOT POCKET

Breathing Life into
Artificial Eyes.



Every AI Avatar Today Gets Eyes Wrong

The uncanny valley is built on three distinct failure states.

Bust 1 (The Creepy Stare)



Eye Contact Correction

Systems like Apple FaceTime, NVIDIA, and Microsoft Teams over-correct, creating a constant, locked-on stare that induces deep discomfort.

Bust 2 (The Random Swim)



Procedural Systems

Tools like Audio2Face and D-ID move eyes randomly. Studies prove random gaze is psychologically worse than no movement at all.

Bust 3 (The Disconnect)



Zero Cognitive Link

No system connects visual gaze detection to conversational intelligence, resulting in dead eyes, broken trust, and zero human connection.

Why Eyes Matter More Than Anything Else.

Eye contact triggers a neurochemical cascade no other cue can match.

Oxytocin Release

The bonding hormone spikes.
(Source: Nagasawa 2015)

Amygdala Activation

Instant social significance
detection.

Prefrontal Cortex Sync

Brains literally synchronize
during mutual gaze.
(Source: Hirsch 2017)

Mirror Neuron Engagement

The fundamental neural basis
for feeling seen.
(Source: Frontiers in Psychology 2018)

Dopaminergic Reward

The brain provides a chemical
reward for sustained,
appropriate eye contact.



The Eye Contact Paradox.

Connection comes from facial gaze, not eye-locking.

3.0
SECONDS

Perception:

The perceived duration of sustained eye contact.

0.36 SECONDS

Reality:

Actual mutual eye contact per episode (only 10% of total conversation time).

The Mathematical Rules of Gaze

Speaker

Looks at listener
60-70% of the time.
Ends 71% of turns with
direct gaze (turn-yielding).

Listener

Looks at speaker
70-80% of the time.

Aversion

30-40% of conversation
time must be spent
looking AWAY.

Latency

200-400ms natural
response delay when
meeting gaze.

Source: Binetti 2016
(N=498, 56 nations)

The Industry Gap: Beautiful Renders, Dead Souls.

The sensing tech exists. The animation tech exists.
Nobody has built the intelligence layer.

Gaze Intelligence
(Random/Fixed vs. Cognitively Motivated)



D-ID

Pre-rendered /
Random. Eyes
swim, noticeably
synthetic.



Synthesia, HeyGen

Forward stare,
no interactivity.



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Autonomous +
Cognitively Motivated.



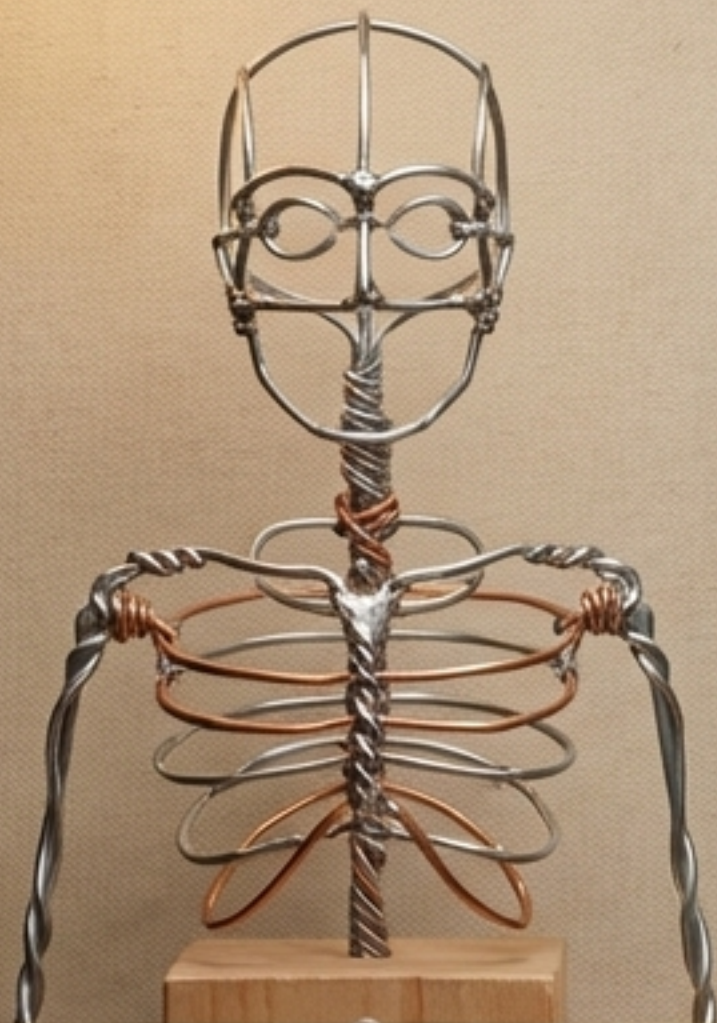
Apple FaceTime, Teams,
NVIDIA ACE, MetaHuman

Temporal flickering,
procedural saccades, or
zero autonomous gaze.

Interactivity
(Pre-rendered vs. Autonomous)

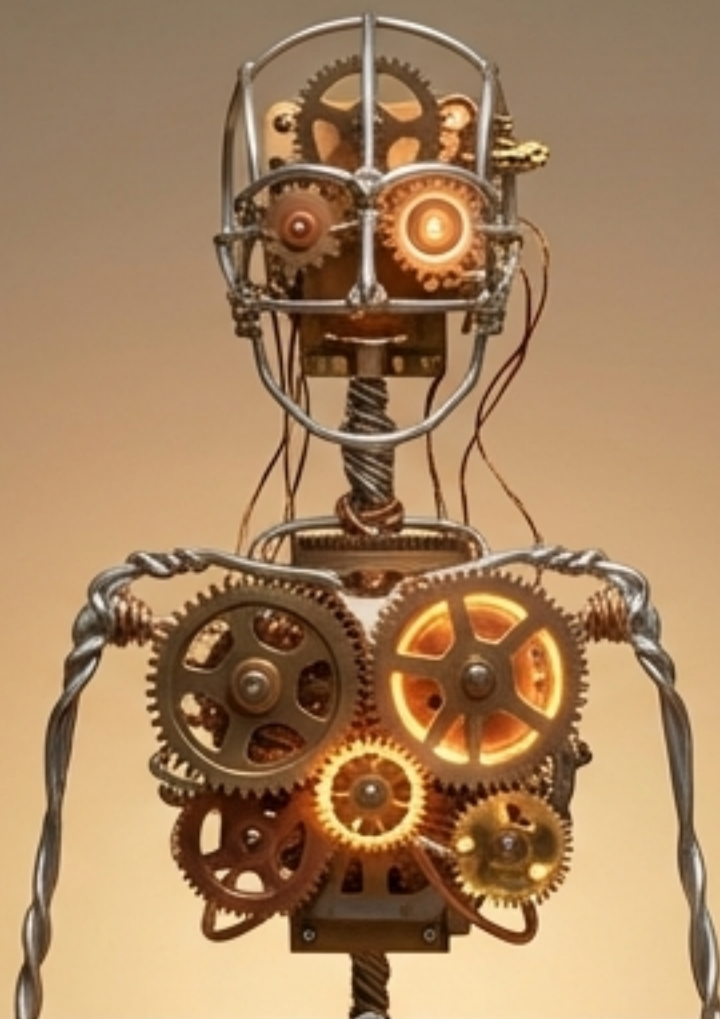
The Armature of Human Connection.

Thot Pocket's 3-Layer Architecture.



LAYER 1: SENSE

User gaze detection via MediaPipe / L2CS-Net. Classifies zones (at me, away, down). Tracks blinks and head poses.



LAYER 2: THINK

The Conversational State Machine. Culturally-tuned gaze allocation. Cognitively-motivated aversion. Turn-taking signals.



LAYER 3: ACT

Procedural 60Hz animation engine. Outputs quaternions, pupil size, and blink triggers driving saccades, VOR, and vergence.

The 'Think' Engine: Gaze with Intent.

Translating conversation states
into cognitive eye movement.

LISTENING

70-80% gaze at user.
Aversion at 20-30%.
Blinks reduced to 12/min.



THINKING

30-50% gaze at user.
Looks up-right (constructing)
or down-left (deliberating).



IDLE

30-40% gaze at user.
Slow environmental scanning.



SPEAKING

60-70% gaze at user.
Blinks elevated to 22/min
at exact phrase boundaries.

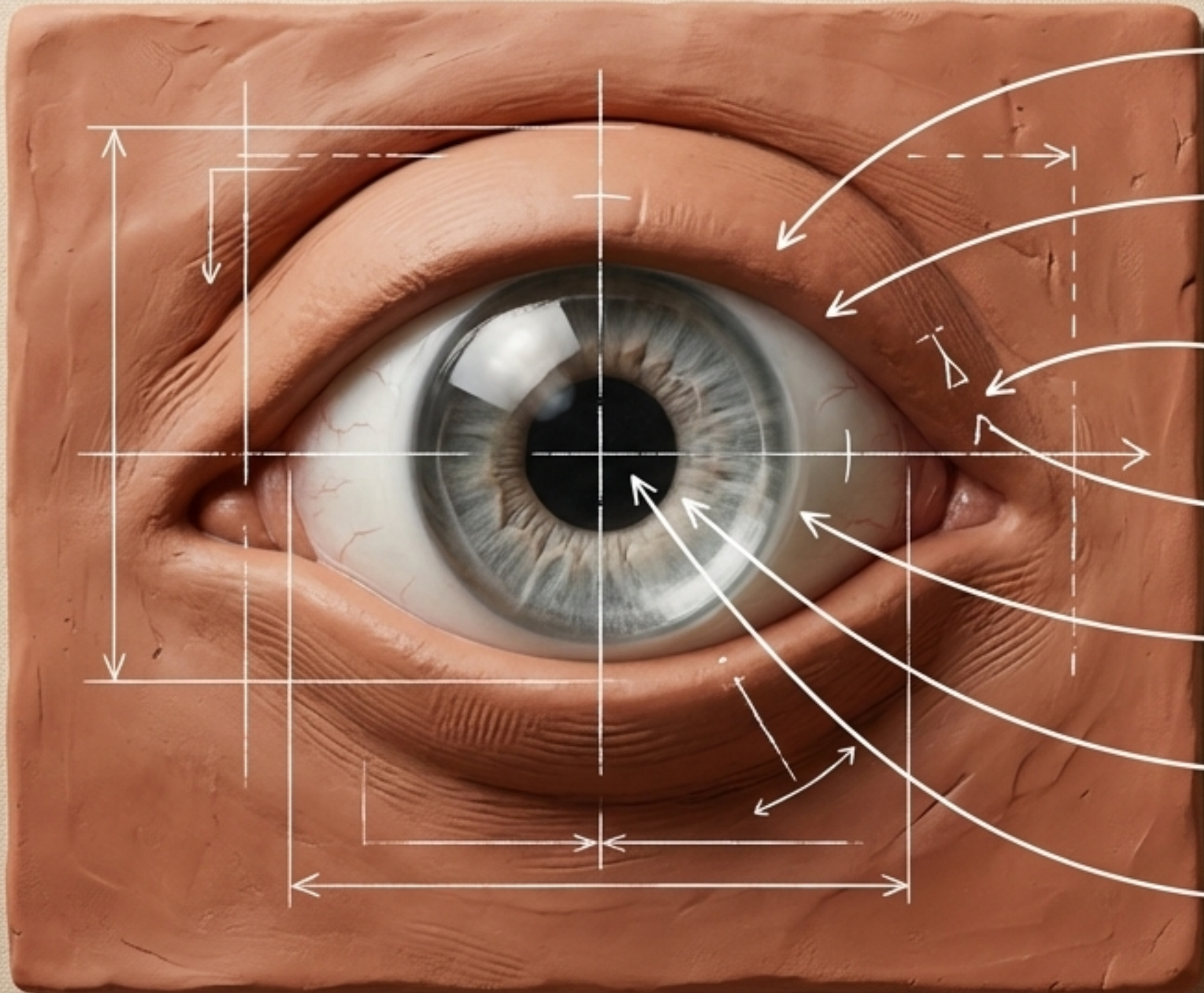


Core Rules:

No single eye contact episode exceeds 3.2s without a micro-break.
Avatar finishes speaking: establish eye contact (turn yield).
Reduce eye contact when delivering disagreeable information (Chen 2013).

What Turns 'Dead' into 'Alive'?

The mechanical nuances missing from current procedural avatars.



Micro-saccades: 0.2-0.5 degrees at 1-2Hz.
Eyes are NEVER entirely still.

VOR (Vestibulo-ocular Reflex): Eyes counter-rotate precisely during head movement. (Missing this makes eyes look painted on).

Blink-Speech Coupling: Blinks occur naturally at phrase boundaries, never at fixed, robotic intervals.

Triangle Scanning:
Left eye -> Right eye -> Mouth at 2-4Hz.

Pupil Dilation: Fluctuates dynamically with light, cognitive load, and emotion.

Vergence: Each eye moves slightly independently to focus on depth.

Ballistic Saccades: Real eyes snap instantly; they do not linearly interpolate or slide.



One Gaze Does Not Fit All

Culturally calibrated algorithms for global deployment.

One monolithic gaze model fails 70% of the world.



Western (US/Europe):
High contact (60-70%).
Direct gaze signals confidence.



East Asian (Japan/Korea/China):
Lower contact (30-50%).
Prolonged gaze is perceived as rude.



Middle Eastern:
Very high same-gender contact. Cross-gender requires specific aversion logic.



South Asian:
Lower contact with authority figures. Avoidance equals respect.

Thot Pocket dynamically shifts frequency, duration, aversion patterns, and intensity through a single API culture parameter.

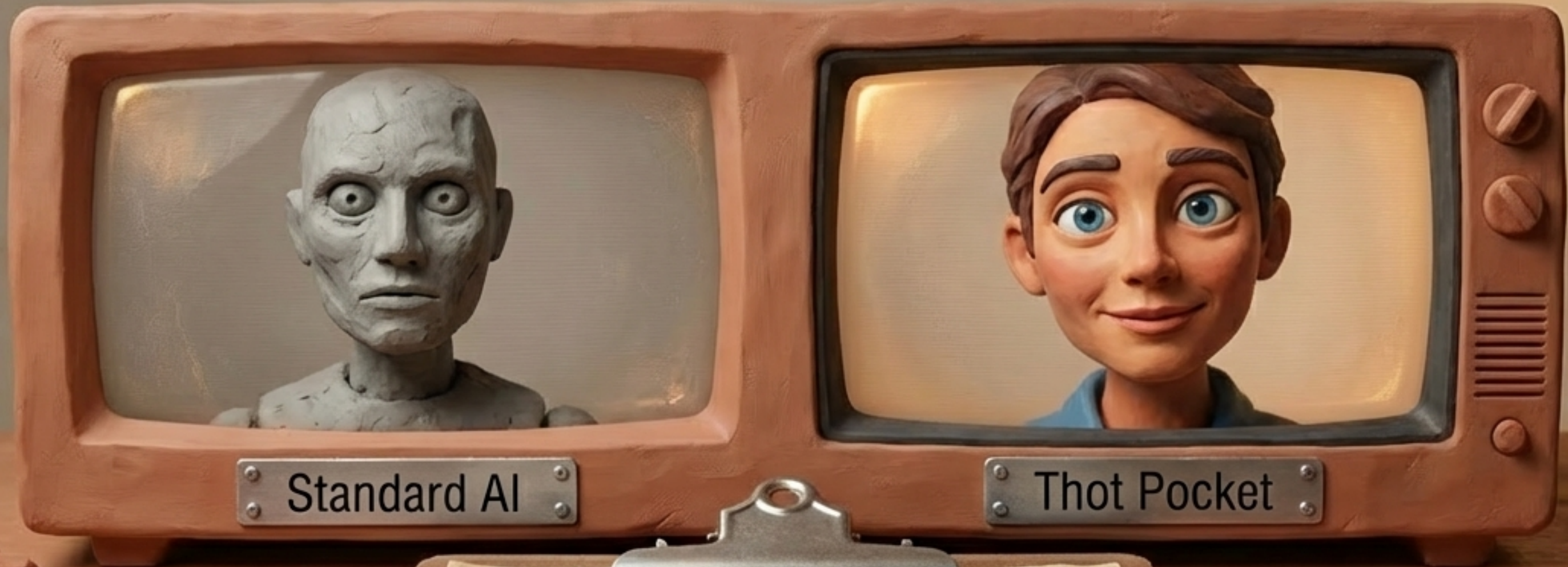
Built on Science, Not Guesswork.

The foundation of 25+ peer-reviewed sources.



Feel the Difference Before You Analyze It.

Side-by-side behavioral comparison.



Standard AI

Thot Pocket

Demo Scenarios:

- Scenario 1:** User looks at avatar; avatar meets gaze with natural 300ms delay.
- Scenario 2:** Avatar "thinks" before responding, initiating a cognitively-motivated look-away.
- Scenario 3:** Avatar finishes speaking, immediately establishing eye contact as a turn-yield.
- Scenario 4:** Sustained conversation highlighting the natural rhythm of contact and aversion.

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The Intelligence Layer of Human Connection



The sensing technology exists.

The rendering technology exists.

Thot Pocket is the intelligence
that connects them.